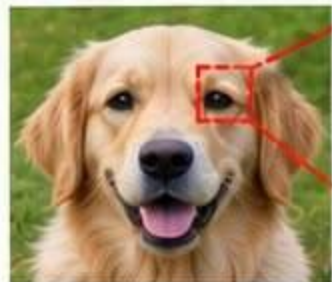


Computer Vision every data analyst should know

1. Image Representation

A digital image is represented as an array of pixel values.



176	180	178
177	175	173
170	168	169

Image
(H × W × C)

Pixel values
(grayscale)

- Grayscale image: H × W (1 channel)
- Color image: H × W × 3 (R, G, B channels)

2. Image Preprocessing

We enhance images to improve model performance.



Noisy Image

Denoising



Denoised Image



Original
(224 × 224)

Resizing



(112 × 112)

- Resize
- Normalize
- Denoise
- Convert to grayscale
- Data augmentation

3. Convolution Operation

A filter (kernel) slides over the image to extract important features.

1	1	1	0	0
1	1	1	0	0
1	1	1	0	0
0	0	0	1	1
0	0	0	1	1

Image

1	0	-1
1	0	-1
1	0	-1

Filter
(Kernel)

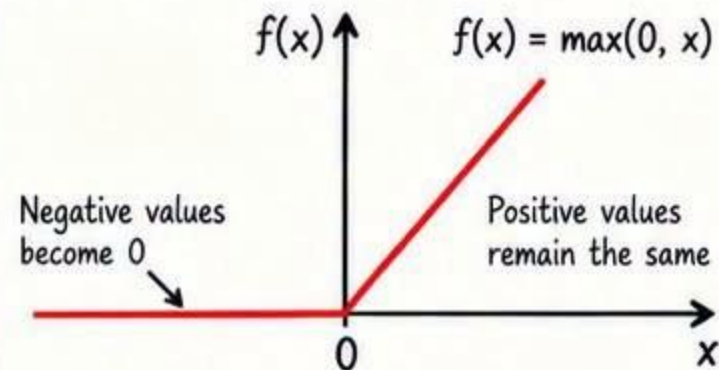
0	1	1
0	1	1
0	1	1

Feature Map

Convolutions detect edges, textures and patterns in the image.

4. Activation Function (ReLU)

Introduces non-linearity by setting negative values to zero.



5. Pooling

Reduces the spatial size and keeps the most important information.

Input (4 × 4)

1	2	2	0
0	1	3	1
2	0	1	0
1	3	2	1

Max Pooling
(2 × 2)

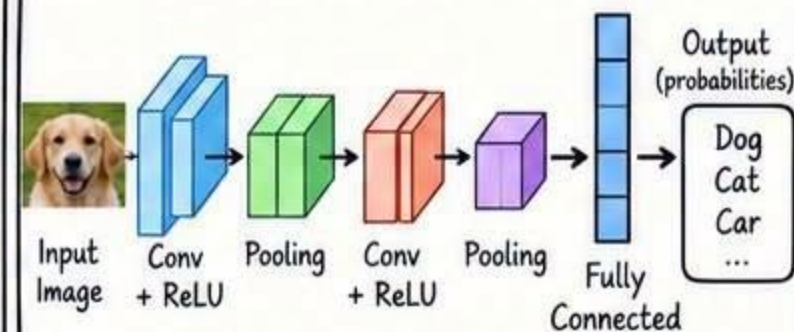
Output (2 × 2)

2	3
3	2

Pooling helps reduce computation and makes the model more robust to small changes.

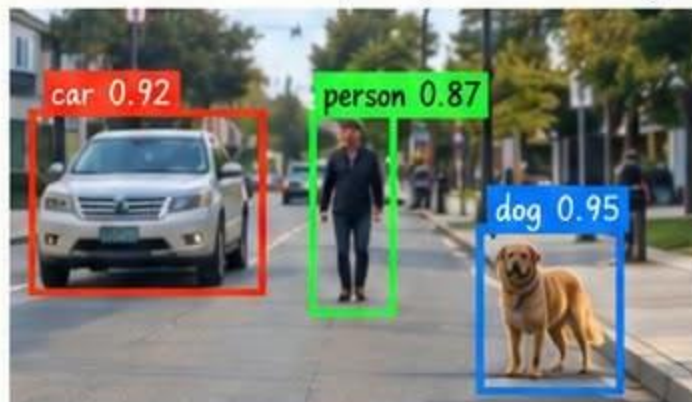
6. CNN Architecture

A Convolutional Neural Network (CNN) stacks multiple layers to learn complex features.



7. Object Detection

Finds and locates multiple objects in an image.



Outputs: class label + bounding box coordinates

8. Image Segmentation

Assigns a class label to each pixel in the image.



Original Image



Semantic Segmentation

- | | | |
|--------|----------|-----|
| Road | Sidewalk | Car |
| Person | Tree | Sky |

9. Applications of Computer Vision

Used in many real-world applications.



Self-driving cars



Medical imaging



Face recognition



Industrial inspection



Agriculture monitoring



Video surveillance



Image search & retrieval



Retail & E-commerce

Computer Vision Explained

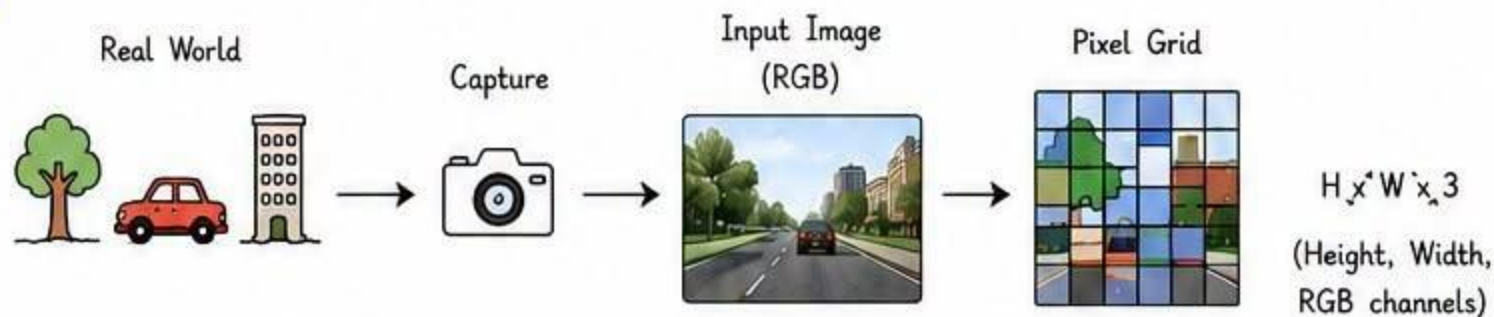
From pixels to understanding the visual world.

□ = Data / Feature
→ = Flow

1. Image Acquisition

Capture images from the real world.

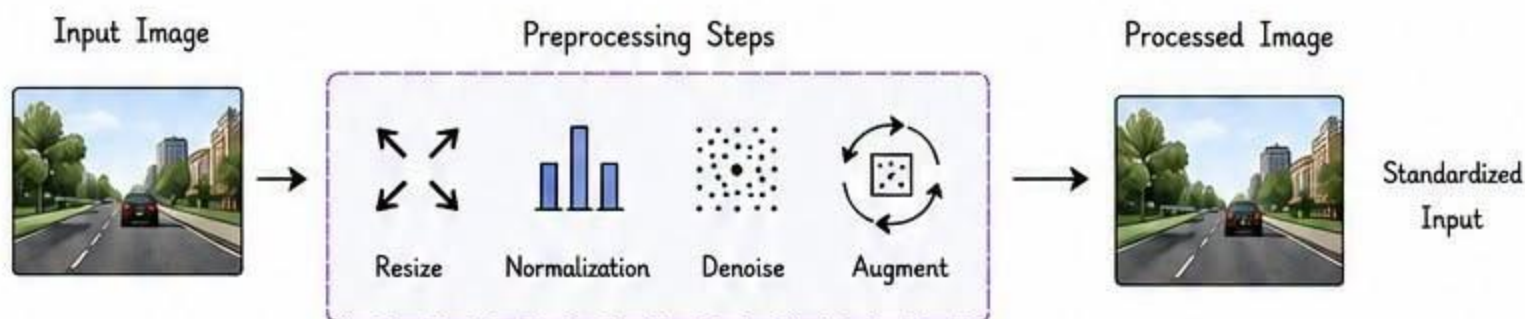
- Cameras, sensors, datasets
- Raw pixel data



2. Preprocessing

Prepare the image to improve learning and performance.

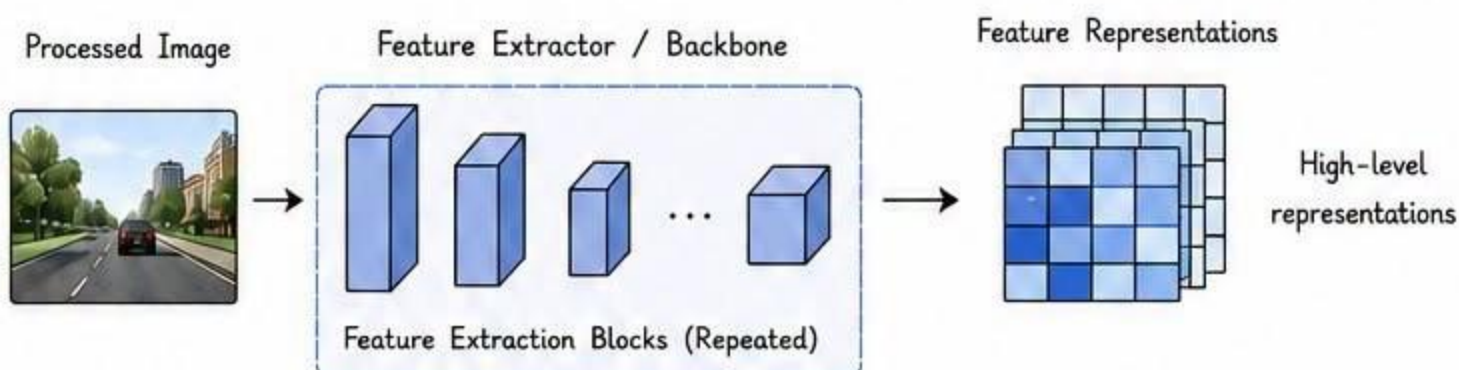
- Resize, normalization, denoise
- Augmentation (optional)



3. Feature Extraction

Extract meaningful features from pixels.

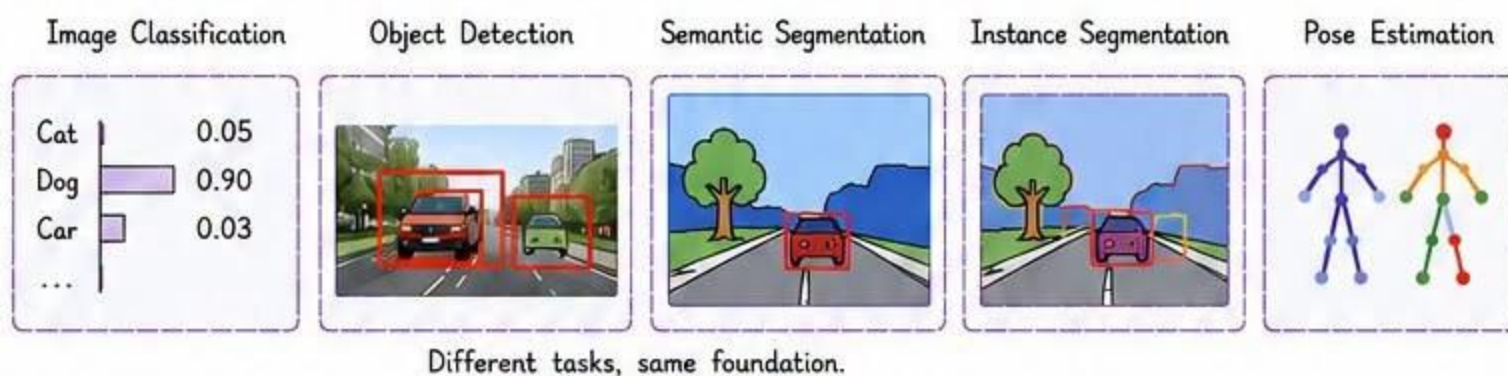
- Learn visual patterns and representations
- Learned automatically by the model



4. Understanding (Tasks)

Use features to perform computer vision tasks.

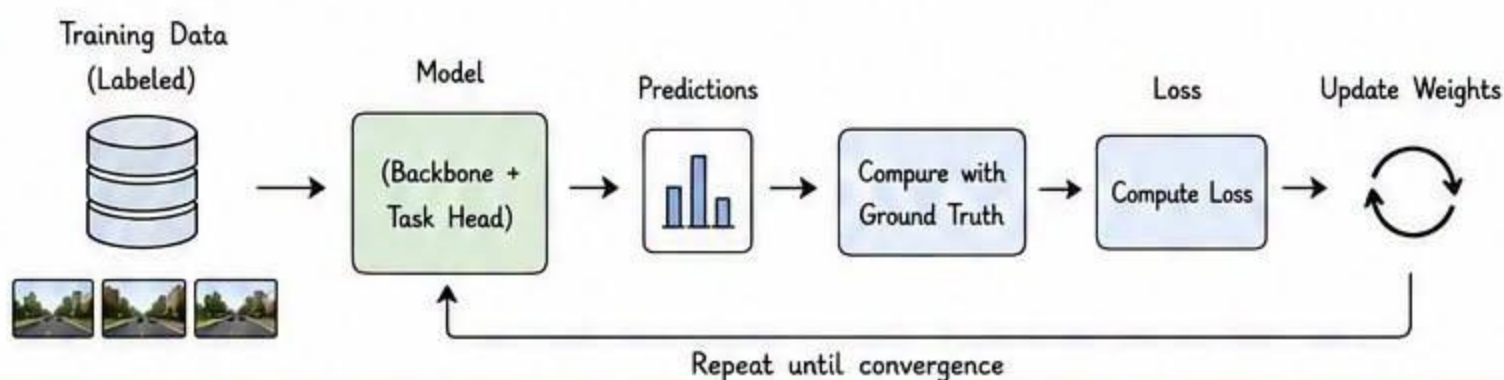
- Many possible tasks
- Task-specific output heads



5. Learning (Training)

Model learns patterns from data.

- Training data
- Loss calculation
- Backpropagation & optimization



6. Inference (Prediction)

Use trained model to make predictions on new images.

- New, unseen images
- Forward pass

